

Education

University of Michigan, Ann Arbor	<i>Ph.D., Computer Science & Engineering</i>	2027 (expected)
◦ GPA: 3.97/4.0 ◦ Topics: Computer Architecture, Autonomous Systems, Sustainable Computing ◦ Relevant Coursework: Advanced Compilers, Microarchitecture, Parallel Computer Architecture		
Addis Ababa University, Ethiopia	<i>B.Sc., Electrical & Computer Engineering</i>	2020

Experience

Architecture Performance Modeling Intern , <i>imec</i>	<i>Ann Arbor, MI</i>	2025
◦ End-to-end profiling of Autoware autonomous driving's perception pipeline (LiDAR, radar, camera, GNSS/IMU) across heterogeneous platforms (CPU/GPU/edge SoCs). ◦ Built a modular toolkit for per-node latency extraction, DAG-based dependency graphs, and critical-path analysis. Used Intel VTune and ROS 2 CARET tracing.		
Undergraduate Research Intern , <i>University of Michigan</i>	<i>Remote</i>	2021
◦ BeXAI: Built benchmark suite for explainable AI with curated models and fidelity evaluation metrics. Github repo ◦ Graph acceleration: Evaluated reordering techniques for graphs on baseline and custom accelerator. Github repo		

Publications

Evergreen: Comprehensive Carbon Modeling for Emission-Performance Tradeoffs [<i>IISWC'24</i>]	Sept 2024
10.1109/IISWC63097.2024.00021	
<i>Tersiteab Adem</i> , Andrew McCrabb, Vidushi Goyal, Valeria Bertacco	
Octal: Efficient Automatic Data-Oblivious Transformations to Eliminate Side-Channel Leakage [<i>SecDev'24</i>]	Oct 2024
10.1109/SecDev61143.2024.00018	
Biniyam Tiruye, Lauren Biernacki, <i>Tersiteab Adem</i> , Petros Mola, Todd Austin	

Projects

Research Projects

- **Mnemosyne: Temporal Redundancy for Efficient LiDAR 3D Object Detection** (*Under Submission*)
Developed a speculative memoization framework that uses a lightweight inter-frame similarity checker and motion compensation to reuse detection outputs or intermediate activations across consecutive LiDAR frames. Evaluated five model-dataset benchmarks, achieving $5.8\times$ average speedup (up to $10.2\times$) and $2.39\times$ energy reduction with approximately 10% accuracy loss.
- **AutoProbe: Characterizing and Profiling ADAS/AD Workloads in Autoware** (*Under Submission*)
Built a multi-level profiling framework that combines ROS 2 pipeline tracing, LLVM-based architecture-agnostic instrumentation, and hardware-specific binary profiling. Reconstructed Autoware task graphs to measure per-node and end-to-end latency, communication behavior, critical paths, instruction mix, memory traffic, and roofline characteristics across heterogeneous platforms and sensor configurations.

Course Projects

- **COCO: Compiler-Based Obliviousness Checker and Optimizer:** Developed a taint-analysis-based compiler framework to verify data obliviousness of workloads, ensuring memory access patterns are independent of input data.
- **ACRE: Explainable Random Forest Process-in-Memory Accelerator:** Parallelized explainable random forest inference using a Process-in-Memory (PiM) accelerator by offloading comparison operations into memory, to reduce off-chip memory traffic.
- **Parallel Simulation of Coarse-Grained Reconfigurable Arrays using SST:** Implemented a CGRA-based systolic array for GeMM kernels in the Structural Simulation Toolkit, achieving $5\text{--}24\times$ speedup over RTL simulation.
- **Memory Access Pattern Recognition using Search Algorithms:** Applied hill climbing, simulated annealing, and genetic algorithms to infer memory access patterns from workload address traces to inform memory prefetching.

Technical Skills

Languages: Python, C/C++, CUDA, Java, C#, Bash, Makefile **Compilers:** Clang, LLVM **Verification:** Murphi
Robotics: ROS2, Autoware **Profilers:** NSight Compute/Systems, Torch Profiler, Intel VTune, Intel Pin, NVBit
Simulation Tools: gem5, SST **ML Libraries:** TensorFlow, PyTorch